

# SARAVANAN RAMESH

Full Stack Engineer | React.js · Next.js · Node.js · TypeScript · MySQL · MongoDB · AWS

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## SUMMARY

Full Stack Engineer with 4+ years building and shipping multi-tenant SaaS products end to end — LMS, CRM, ERP/POS, website-builder and marketplace platforms on React.js, Next.js, Node.js, MySQL, MongoDB and AWS. Delivered 10+ production applications covering RBAC and tenant isolation, subscription billing, real-time communication, background job processing and event-driven serverless pipelines, cutting manual processing effort by 90% and API response times by up to 60%. Also owns the AI layer on these products: production RAG and semantic search over a 150+ table schema using OpenAI, Gemini and Claude, and a from-scratch PyTorch CNN that automates exam-sheet evaluation. Comfortable from database schema design through to deployment on self-managed AWS infrastructure.

## CORE SKILLS

**Frontend:** React.js, Next.js, TypeScript, JavaScript (ES6+), Redux Toolkit, Zustand, Tailwind CSS, Vue.js, Ant Design

**Backend:** Node.js, Express.js, REST API design, WebSockets / Socket.IO, background jobs, Laravel, PHP, Swagger / OpenAPI

**Databases:** MySQL, MongoDB, Mongoose, Knex, schema design, indexing, query optimization, cursor-based pagination

**Cloud & DevOps:** AWS Lambda, S3, SQS, EC2, CloudFront · serverless and event-driven architecture · Nginx, PM2, Let's Encrypt

**AI / ML:** OpenAI, Anthropic Claude, Google Gemini, RAG pipelines, Qdrant vector DB, semantic search, PyTorch (CNN), llama.cpp, AssemblyAI

**Security:** JWT, OAuth 2.0, RS256, RBAC, multi-tenant isolation, rate limiting, XSS and NoSQL injection prevention

**Integrations:** Stripe, Razorpay, Square, Zoom SDK, Agora, Twilio, Firebase / FCM, Google Calendar, Cloudinary

## PROFESSIONAL EXPERIENCE

**GNXTACE Technology** — Full Stack Developer | Chennai, India | May 2024 – Present

**NexusOne.coach** — Multi-Tenant LMS and Live Coaching Platform · React.js, Node.js, MySQL, Socket.IO, Stripe, Zoom SDK, OpenAI/Gemini/Claude

[portal.nexusone.coach](https://portal.nexusone.coach) · [student.nexusone.coach](https://student.nexusone.coach)

- Built the platform end to end across **five integrated production workflows** — courses, assessments, live classes, cohorts and instructor payouts — delivered through student, admin and super-admin frontends.
- Added an **AI learning layer**: transcript-grounded Q&A and semantic search over a **150+ table MySQL schema**, integrating OpenAI, Gemini and Claude behind a Qdrant vector store, with an ffmpeg/play-dl media ingestion and PDF-parsing pipeline feeding the index.
- Secured tenant operations with RBAC authorization, session controls and administrative activity logging enforced across every protected action.
- Automated the full live-class lifecycle by integrating Zoom SDK, Google Calendar, Stripe and Firebase Cloud Messaging, removing manual scheduling, payment and notification handoffs for tenant administrators.

**Seltrix** — Multi-Tenant Website Builder and Template Marketplace · Next.js, Node.js, MySQL, Stripe, Cloudinary | [seltrix.live](https://seltrix.live)

- Built multi-tenant website publishing with custom domains, template commerce and public tenant rendering using dynamic Next.js pages, Node.js APIs and Stripe checkout workflows.
- Created a schema-driven template authoring pipeline with automated validation, live preview and catalog import, so new marketplace templates ship without code changes.
- Built super-admin and tenant-admin panels for company setup, template assignment and reporting, with filtering, pagination and role-scoped views.

**ZPONZ** — Expert Marketplace, Wallet and CMS Platform · React.js, Node.js, MySQL, Socket.IO, Stripe, AWS

- Built Stripe-backed wallet services supporting deposits, withdrawals, transfers, escrow balances and scheduled expert payouts.
- Expanded real-time engagement with Socket.IO and FCM notifications, and served cloud assets through an S3/CloudFront distribution pipeline.
- Enabled live consultations and rich media handling by integrating Agora and Twilio with an ffmpeg/Sharp/Jimp media pipeline, plus an admin CMS for platform-wide content moderation.

**Jnana Inventive Pvt. Ltd.** — Full Stack Developer | Chennai, India | June 2022 – May 2024

**AdUnity** — Multi-Tenant CRM and AI Call Analytics Platform · React.js, Node.js, SQL, Socket.IO, AssemblyAI

- Built a tenant-isolated real-estate CRM covering lead tracking, site visits, team analytics and lead expiry across React.js interfaces, Node.js APIs and SQL data models.
- Automated call intelligence — click-to-call, transcription, sentiment analysis and live incoming-call alerts — by integrating Aloha, Knowlarity, AssemblyAI and Socket.IO.
- Replaced manual operational reporting with scheduled report generation, Excel/PDF exports and FullCalendar-driven visit scheduling across role-based dashboards.

**QuickBizz** — Multi-Tenant ERP/POS SaaS Platform · Laravel, Vue.js, MySQL, Stripe, Razorpay | [pos.quickbizz.io/admin/login](https://pos.quickbizz.io/admin/login)

- Built a company-isolated ERP/POS platform covering billing, inventory, purchasing, warehouses, storefronts and PDF invoicing — **six production workflows** on Laravel, Vue.js and MySQL.
- Automated subscription governance across **15 functional modules** with usage-based access control and overdue-account restrictions via Stripe and Razorpay.
- Built point-of-sale operations with barcode scanning and generation, QR-based lookups, and PDF invoice generation via jsPDF and html2pdf.

## INDEPENDENT PROJECT

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**Appolo Smart Test** — Enterprise Exam Evaluation Platform | Ongoing

[appolosmarttest.com](https://appolosmarttest.com) · [admission.appolosmarttest.com](https://admission.appolosmarttest.com) · [iOS app](#) · [Android app](#) | React.js · Node.js · MongoDB · PyTorch · AWS Lambda, S3, SQS · Firebase · TUS

- Built a **convolutional neural network from scratch in PyTorch to evaluate OMR answer sheets**, scoring scanned submissions automatically and removing manual paper correction from the exam workflow entirely.
- **Cut manual PDF-processing effort by 90%** by designing a serverless ingestion pipeline on AWS Lambda, S3 and SQS.
- **Improved API response times by up to 60%** through MongoDB indexing, lean queries, cursor-based pagination and reusable Mongoose plugins.
- Built **50+ production REST endpoints** in Node.js with RS256 JWT authentication, request validation and structured logging, documented for client consumption.
- Delivered a **40+ route** responsive React admin experience with reusable screens and optimized state management, alongside published student apps on iOS and Android.
- Stabilized **1,000+ concurrent push notifications** without memory spikes using an FCM and SQS queue-worker architecture, and enabled **resumable uploads exceeding 500 MB** via the TUS protocol with retry-safe handling and security controls.

## ADDITIONAL PROJECTS

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**VoxIcons** — AI Chat Assistant Platform · React.js, Node.js, Socket.IO, OpenAI, node-llama-cpp

- Built a dual-mode AI chat assistant with real-time streaming conversations backed by both the OpenAI API and a **self-hosted node-llama-cpp inference service**, with Socket.IO messaging and a wallet-based usage and credit system.

**Ghana Visa** — Visa Application and Payment Platform · React.js, Node.js, MySQL, AWS S3, Stripe, Square

- Built an end-to-end visa application workflow with document upload, staff review and application-status logging, using AWS S3 document storage and dual payment gateways.

**Journal** — Real-Time Blogging and CMS Platform · Next.js, Node.js, MySQL, Socket.IO, Swagger

- Built a real-time content platform with live-updating blog feeds and a fully documented REST API, using a Next.js reader app, a Node.js/Knex backend and Socket.IO-driven updates.

**Portfolio** — Live System-Design Showcase · Next.js, Express, SQLite, AWS EC2, Nginx, PM2 | [saravanan.krelyvo.in](https://saravanan.krelyvo.in)

- Self-hosted and deployed on an AWS EC2 instance with Nginx reverse proxy, PM2 process management and HTTPS via Let's Encrypt — every architectural claim on the site is backed by a live, interactive backend API.

## EDUCATION & TRAINING

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**B.Sc. Electronics and Communication** — SRM Arts & Science College, Chennai | 2018–2021 | CGPA 7.8/10

**Self-directed full-stack training**, 2021–2022 — JavaScript, React and Node.js fundamentals, completed prior to entering the industry.